

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 5 — Chemical Energetics

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	5.1 Enthalpy Change · Standard Conditions · Calorimetry · 5.2 Hess's Law · Bond Energies
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	221 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

5.1 Enthalpy change, ΔH
5.2 Hess's law

- Standard conditions (298 K, 100 kPa), definitions of formation, combustion, neutralisation, and atomisation
- Conservation of energy in cycles, calculations using standard enthalpies of formation and combustion

SUBTOPIC 5.1 — ENTHALPY CHANGE, ΔH & CALORIMETRY

1 Standard Enthalpy of Formation & Standard Conditions — 9701/22/M/J/21/Q3(a)–(b)

EASY · Syllabus 5.1

Thermochemical equations depend on precisely defined standard states and conditions.

- (a) Define the term standard enthalpy change of formation, ΔH_f^θ .

[3]

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- (b) State the standard conditions of temperature and pressure specified for thermochemical measurements.

[2]

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- (c) Explain why the standard enthalpy change of formation of pure liquid bromine, $\text{Br}_2(\text{l})$, is exactly 0.0 kJ mol^{-1} .

[1]

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2 Enthalpy Profile Diagrams: Catalysed vs Uncatalysed — 9701/21/O/N/20/Q3(a)–(c)

EASY · Syllabus 5.1

The enthalpy profile diagrams below show the progress of an exothermic reaction and an endothermic reaction, with and without a catalyst.

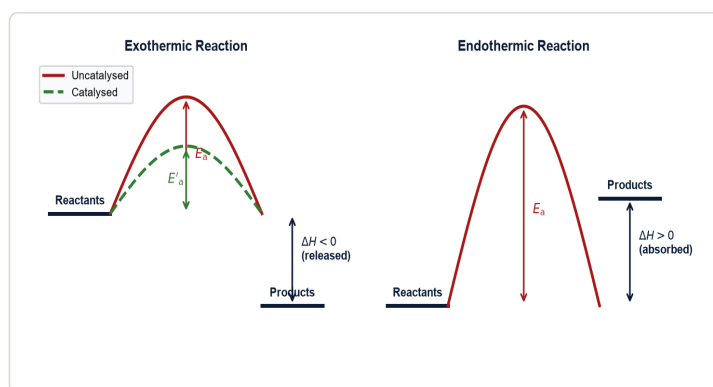


Fig. 5.1: Enthalpy profile diagrams for exothermic and endothermic reactions.

- (a) Define the term activation energy, E_a .

[1]

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- (b) With reference to Fig. 5.1, explain how a catalyst increases the rate of an exothermic reaction without altering ΔH .

[2]

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- (c) State the mathematical relationship between the activation energy of the forward reaction, $E_a(\text{forward})$, the activation energy of the reverse reaction, $E_a(\text{reverse})$, and ΔH for an endothermic reaction.

[1]

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3 Standard Enthalpy of Combustion Definition & Equation — 9701/23/M/J/22/Q2(a)–(b)

EASY · Syllabus 5.1

Combustion reactions are invariably exothermic processes.

- (a) Define the term standard enthalpy change of combustion, ΔH_{c} . [3]

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- (b) Write the thermochemical equation representing the standard enthalpy change of combustion of liquid ethanol, $\text{C}_2\text{H}_5\text{OH}(\text{l})$, including state symbols. [2]

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4 Standard Enthalpy of Neutralisation & Weak Acids — 9701/22/F/M/21/Q4(a)–(c)

HARD · Syllabus 5.1

The standard enthalpy change of neutralisation between strong hydrochloric acid and strong sodium hydroxide is $-57.1 \text{ kJ mol}^{-1}$.
 $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \quad \Delta H = -57.1 \text{ kJ mol}^{-1}$

- (a) Define the term standard enthalpy change of neutralisation. [2]

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- (b) Write the ionic equation for the neutralisation reaction between any strong acid and any strong base. [1]

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- (c) When ethanoic acid, CH_3COOH , is neutralised by $\text{NaOH}(\text{aq})$, the experimental enthalpy change of neutralisation is $-55.2 \text{ kJ mol}^{-1}$. Explain why this value is less exothermic than $-57.1 \text{ kJ mol}^{-1}$. [2]

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5 Flame Calorimetry: Enthalpy of Combustion of Propan-1-ol — 9701/21/M/J/22/Q3(a)–(c)

HARD · Syllabus 5.1

A student uses the spirit burner calorimeter apparatus shown in Fig. 5.2 to determine the enthalpy change of combustion of propan-1-ol, $\text{C}_3\text{H}_7\text{OH}$ ($M_r = 60.0$).

- Mass of water in copper calorimeter = 150.0 g
 - Initial mass of burner + propan-1-ol = 78.45 g
 - Final mass of burner + propan-1-ol = 77.85 g
 - Initial temperature of water = 19.5 °C
 - Final temperature of water = 44.0 °C
- (Specific heat capacity of water, $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$)

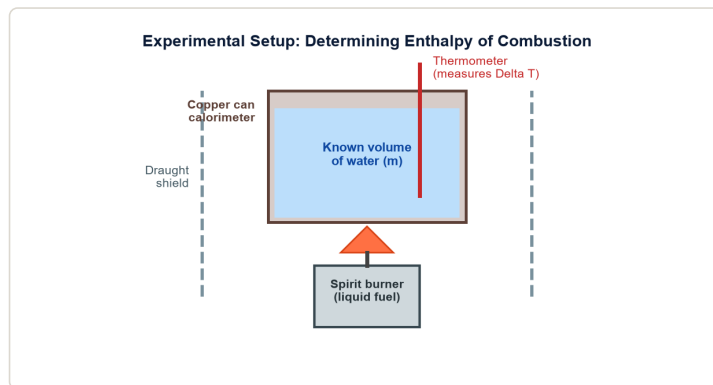


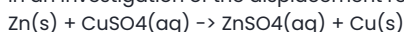
Fig. 5.2: Flame calorimeter apparatus for measuring enthalpy of combustion.

- (a) Calculate the heat energy, q in kJ, absorbed by the water. [2]
- (b) Calculate the amount, in moles, of propan-1-ol burned and hence determine the experimental ΔH_c in kJ mol^{-1} . [2]
- (c) The data book value for ΔH_c of propan-1-ol is $-2021 \text{ kJ mol}^{-1}$. State two major experimental sources of error that explain why the student's value is significantly less exothermic. [2]

6 Calorimetry Cooling Curve & Graphical Extrapolation — 9701/22/O/N/23/Q2(a)–(c)

HARD · Syllabus 5.1

In an investigation of the displacement reaction between zinc and copper(II) sulfate:



50.0 cm³ of 1.00 mol dm⁻³ CuSO₄ was placed in an expanded polystyrene cup. Zinc powder (excess) was added at exactly $t = 3.0$ minutes. The temperature–time cooling curve is shown below.

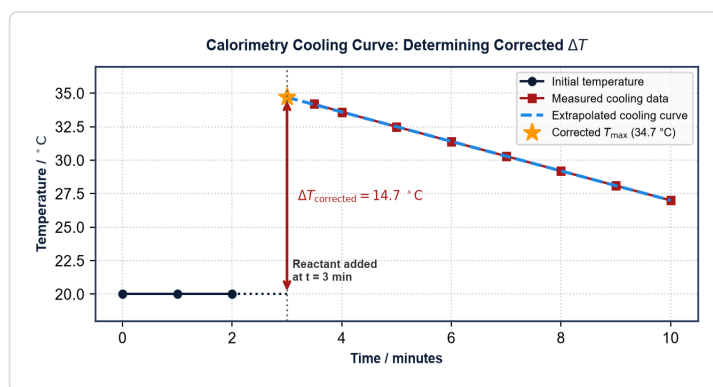


Fig. 5.3: Temperature–time cooling curve for the $\text{Zn} + \text{CuSO}_4$ displacement reaction.

- (a) Using Fig. 5.3, determine the corrected maximum temperature change, $\Delta T_{\text{corrected}}$, for this reaction. [2]

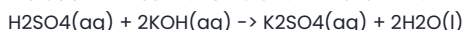
- (b) Calculate the enthalpy change, ΔH , in kJ mol^{-1} , for the displacement reaction. (Assume solution density = 1.00 g cm^{-3} , $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$). [3]
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- (c) Explain why an expanded polystyrene cup is used as a calorimeter rather than a glass beaker. [1]
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7 Enthalpy of Neutralisation: Sulfuric Acid vs Potassium Hydroxide — 9701/21/M/J/23/Q3(a)–(b)

HARD · Syllabus 5.1

A student mixes 25.0 cm^3 of $1.00 \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4$ with 50.0 cm^3 of $1.00 \text{ mol dm}^{-3} \text{ KOH}$ in an insulated cup.



The initial temperature of both solutions was 21.0°C . The maximum recorded temperature was 29.8°C . (Density = 1.00 g cm^{-3} , $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$)

- (a) Calculate the heat evolved, q in Joules, during this reaction. [2]
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- (b) Calculate the standard enthalpy change of neutralisation, ΔH_{neut} , in kJ mol^{-1} of water formed. [2]
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8 Enthalpy of Solution of Ammonium Chloride — 9701/22/M/J/20/Q4(a)–(c)

EASY · Syllabus 5.1

When 5.35 g of solid ammonium chloride, NH_4Cl ($M_r = 53.5$), is dissolved in 100.0 cm^3 of water at 20.0°C , the temperature drops to 16.2°C .

- (a) State whether the dissolution of ammonium chloride is an exothermic or endothermic process. [1]
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- (b) Calculate the enthalpy change of solution, ΔH_{sol} , of NH_4Cl in kJ mol^{-1} . [3]
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- (c) Draw a fully labelled enthalpy profile diagram for this dissolution process showing reactants, products, and ΔH . [2]
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9 Standard Enthalpy of Atomisation Definition & Halogens — 9701/21/O/N/22/Q3(a)-(b)

EASY · Syllabus 5.1

The standard enthalpy change of atomisation involves producing free gaseous atoms.

- (a) Define standard enthalpy change of atomisation, ΔH_{at} . [3]

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- (b) Write the thermochemical equation representing the standard enthalpy change of atomisation of liquid bromine, $\text{Br}_2(\text{l})$. [1]

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- (c) Explain the relationship between the bond dissociation enthalpy of gaseous chlorine, $E(\text{Cl}-\text{Cl})$, and its standard enthalpy of atomisation, $\Delta H_{at}(\text{Cl}_2)$. [2]

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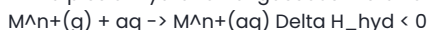
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10 Enthalpy of Hydration & Ion Size — 9701/23/O/N/21/Q3(a)-(b)

HARD · Syllabus 5.1

Enthalpies of hydration of gaseous metal ions are always highly exothermic:



• $\Delta H_{\text{hyd}}(\text{Na}^+) = -406 \text{ kJ mol}^{-1}$

• $\Delta H_{\text{hyd}}(\text{Mg}^{2+}) = -1920 \text{ kJ mol}^{-1}$

- (a) Define the term enthalpy change of hydration. [2]

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- (b) Explain why the enthalpy of hydration of Mg^{2+} is almost five times more exothermic than that of Na^+ . [3]

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11 Calorimeter Heat Capacity Correction — 9701/22/F/M/23/Q2(b)

HARD · Syllabus 5.1

In a more rigorous calorimetry experiment, the heat capacity of the calorimeter container itself must be taken into account:

$$q_{\text{total}} = (m_{\text{water}} \times c_{\text{water}} \times \Delta T) + (C_{\text{calorimeter}} \times \Delta T)$$

- (a) A combustion experiment causes 200.0 g of water inside a copper calorimeter of heat capacity 85.0 J K^{-1} to rise in temperature by 12.4°C . Calculate the total heat energy absorbed, in kJ. [3]

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- (b) Calculate the percentage error that would occur if the heat capacity of the copper container were neglected. [2]

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12 Comparing Combustion Enthalpies in a Homologous Series — 9701/21/M/J/21/Q4(a)–(b)

EASY · Syllabus 5.1

The standard enthalpies of combustion for the first four unbranched alkanes are:

- Methane, CH₄: –890 kJ mol^{–1}
- Ethane, C₂H₆: –1560 kJ mol^{–1}
- Propane, C₃H₈: –2220 kJ mol^{–1}
- Butane, C₄H₁₀: –2877 kJ mol^{–1}

- (a) Describe and explain the trend in standard enthalpies of combustion as the carbon chain length increases. [2]

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- (b) Estimate the standard enthalpy change of combustion of pentane, C₅H₁₂, by calculating the average incremental increase per –CH₂– unit. [2]

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13 Bond Breaking vs Bond Forming Energetics — 9701/22/M/J/22/Q3(a)–(b)

EASY · Syllabus 5.1

Chemical reactions involve the rearrangement of atoms through bond cleavage and bond creation.

- (a) State whether bond breaking is an exothermic or endothermic process and justify your answer in terms of energy changes. [2]

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- (b) Explain why an exothermic reaction results in an overall release of heat to the surroundings in terms of bond energies. [2]

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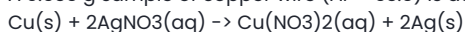
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14 Enthalpy Change of Displacement: Copper and Silver Nitrate — 9701/22/O/N/21/Q4(b)

HARD · Syllabus 5.1

A 0.635 g sample of copper wire (Ar = 63.5) is added to 100.0 cm³ of 0.200 mol dm^{–3} AgNO₃ in an insulated polystyrene beaker:



The temperature of the solution rises by 3.5 °C. (c = 4.18 J g^{–1} K^{–1}, density = 1.00 g cm^{–3})

- (a) Determine whether copper or silver nitrate is the limiting reactant. [2]

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- (b) Calculate the enthalpy change, ΔH , in kJ mol^{-1} of Cu reacted. [3]

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15 Calorimetric Measurement of Neutralisation: Weak Base NH_3 — 9701/21/O/N/23/Q2(b)

HARD · Syllabus 5.1

When 50.0 cm^3 of 1.00 mol dm^{-3} aqueous ammonia is reacted with 50.0 cm^3 of 1.00 mol dm^{-3} HCl in a polystyrene cup, the temperature increases by 6.1°C .

- (a) Calculate the heat released during this reaction in kJ. [2]

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- (b) Calculate the enthalpy change of neutralisation, ΔH_{neut} , in kJ mol^{-1} of NH_4^+ formed. [2]

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- (c) Explain why this ΔH_{neut} value is less exothermic than the reaction between NaOH(aq) and HCl(aq) ($-57.1 \text{ kJ mol}^{-1}$). [2]

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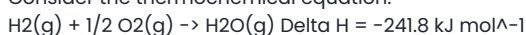
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16 Standard Conditions & Non-Standard State Enthalpies — 9701/23/M/J/20/Q3(a)–(b)

EASY · Syllabus 5.1

Consider the thermochemical equation:



- (a) Explain why $-241.8 \text{ kJ mol}^{-1}$ does NOT represent the standard enthalpy change of formation, $\Delta H_{\text{f}}^\ominus$, of water. [2]

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- (b) Given that the standard enthalpy change of vaporisation of water is $+44.0 \text{ kJ mol}^{-1}$, deduce the true value of $\Delta H_{\text{f}}^\ominus$ for liquid water, $\text{H}_2\text{O(l)}$. [2]

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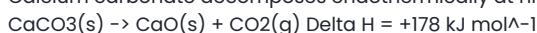
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17 Thermal Decomposition of Limestone: Energetics — 9701/22/F/M/24/Q3(a)–(b)

EASY · Syllabus 5.1

Calcium carbonate decomposes endothermically at high temperatures in an industrial lime kiln:



- (a) State whether heat is absorbed or released by the reaction system during this process. [1]

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- (b) Calculate the thermal energy, in MJ, required to decompose completely 2.50 tonnes of pure CaCO_3 ($M_r = 100.1$). (1 tonne = 10^6 g) [3]
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18 Neutralisation of Diprotic Acid with Monoprotic Base — 9701/21/M/J/24/Q2(b)

HARD · Syllabus 5.1

When 50.0 cm³ of 1.00 mol dm⁻³ H_2SO_4 is mixed with 100.0 cm³ of 1.00 mol dm⁻³ NaOH in a calorimeter, the temperature rises by 9.1 °C.

- (a) Calculate the amount, in moles, of water produced in this reaction. [1]
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- (b) Calculate the heat evolved, q , in kJ. (Total volume = 150.0 cm³, $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$) [2]
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- (c) Calculate the molar enthalpy of neutralisation, ΔH_{neut} , per mole of water formed. [1]
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19 Bond Energy Definitions & Mean Bond Enthalpies — 9701/22/O/N/24/Q3(a)-(b)

EASY · Syllabus 5.1

Bond energies are used to estimate enthalpy changes for reactions involving covalent molecules.

- (a) Define the term mean bond enthalpy. [3]
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- (b) Explain why values of mean bond enthalpies are always positive. [1]
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20 Enthalpy of Combustion of Bioethanol vs Biodiesel — 9701/21/O/N/24/Q2(a)-(b)

HARD · Syllabus 5.1

Bioethanol, $\text{C}_2\text{H}_5\text{OH}$ ($M_r = 46.0$, $\Delta H_{\text{c}} = -1367 \text{ kJ mol}^{-1}$), and methyl octanoate, $\text{C}_9\text{H}_{18}\text{O}_2$ ($M_r = 158.0$, $\Delta H_{\text{c}} = -5480 \text{ kJ mol}^{-1}$), are used as alternative biofuels.

- (a) Calculate the energy released, in kJ, per gram of bioethanol burned. [2]
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- (b) Calculate the energy released, in kJ, per gram of methyl octanoate burned. [1]

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- (c) Suggest one environmental advantage of using biofuels compared to conventional fossil fuels. [1]

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21 Exothermic Solution Process: Dissolution of Anhydrous CaCl₂ — 9701/23/M/J/23/Q3(a)–(b)

EASY · Syllabus 5.1

Anhydrous calcium chloride, CaCl₂, dissolves exothermically in water and is used in self-heating commercial beverage cans. $\text{CaCl}_2(\text{s}) + \text{aq} \rightarrow \text{Ca}^{2+}(\text{aq}) + 2\text{Cl}^{-}(\text{aq})$ $\Delta H_{\text{sol}} = -82.8 \text{ kJ mol}^{-1}$

- (a) Calculate the mass of anhydrous CaCl₂ ($M_r = 111.1$) needed to raise the temperature of 250 cm³ of water by 35.0 °C. ($c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$) [3]

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- (b) Explain why ΔH_{sol} of anhydrous CaCl₂ is exothermic, whereas ΔH_{sol} of NH₄NO₃ is endothermic, in terms of lattice energy and hydration enthalpies. [2]

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22 Combustion Calorimeter: Systematic Errors & Modifications — 9701/22/M/J/25/Q2(b)

EASY · Syllabus 5.1

In a school laboratory combustion experiment using a spirit burner and copper can calorimeter, the experimental enthalpy of combustion of methanol is found to be -480 kJ mol^{-1} . The literature value is -726 kJ mol^{-1} .

- (a) Suggest two specific design modifications to the experimental apparatus that would improve the accuracy of the measured enthalpy change. [2]

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- (b) Explain how incomplete combustion contributes to the lower experimental value. [2]

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23 Direct vs Indirect Determination of Enthalpy Changes — 9701/21/F/M/25/Q3(a)–(b)

HARD · Syllabus 5.1

It is impossible to measure directly the standard enthalpy change of formation of carbon monoxide, CO: $\text{C}(\text{s}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{CO}(\text{g})$ $\Delta H_{\text{f}} = ?$

- (a) Explain why ΔH_{f} of carbon monoxide cannot be determined directly by calorimetry. [2]

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- (b) Describe how Hess's Law allows $\Delta H_f^\ominus$ of carbon monoxide to be determined indirectly. [2]

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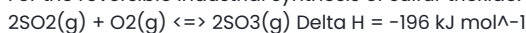
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24 Temperature Dependence of Enthalpy Profiles & Reversibility — 9701/22/O/N/25/Q2(a)-(b)

HARD · Syllabus 5.1

For the reversible industrial synthesis of sulfur trioxide:



- (a) State whether the backward reaction is exothermic or endothermic, and state its ΔH value. [1]

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- (b) Given that the activation energy for the catalysed forward reaction is $+65 \text{ kJ mol}^{-1}$, calculate the activation energy for the catalysed reverse reaction. [2]

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25 Hydration of Copper(II) Sulfate: Indirect Calorimetry — 9701/23/O/N/24/Q3(a)-(b)

HARD · Syllabus 5.1

The direct hydration of anhydrous copper(II) sulfate is too slow to measure directly by calorimetry:



A student measures the enthalpy changes when each solid dissolves in excess water:

- Route 1: $\text{CuSO}_4(\text{s}) + \text{aq} \rightarrow \text{CuSO}_4(\text{aq}) \quad \Delta H_1 = -66.5 \text{ kJ mol}^{-1}$
- Route 2: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s}) + \text{aq} \rightarrow \text{CuSO}_4(\text{aq}) \quad \Delta H_2 = +11.7 \text{ kJ mol}^{-1}$

- (a) Draw or write a Hess's Law cycle connecting $\text{CuSO}_4(\text{s})$, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$, and $\text{CuSO}_4(\text{aq})$. [2]

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- (b) Calculate the enthalpy change of hydration, $\Delta H_{\text{hydration}}$, for anhydrous copper(II) sulfate. [2]

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SUBTOPIC 5.2 — HESS'S LAW & BOND ENERGIES

26 Hess's Law Statement & Thermochemical Cycles — 9701/21/M/J/21/Q5(a)-(b)

EASY · Syllabus 5.2

Hess's Law is a direct manifestation of the fundamental law of conservation of energy.

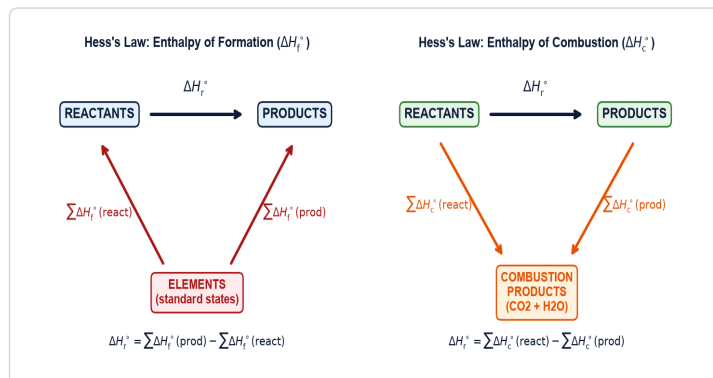


Fig. 5.4: Hess's Law thermochemical cycles using enthalpies of formation and combustion.

- (a) State Hess's Law. [2]

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- (b) With reference to Fig. 5.4, state the mathematical formula used to calculate the enthalpy of reaction ΔH_r from standard enthalpies of formation ΔH_f° . [1]

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- (c) State the mathematical formula used to calculate ΔH_r from standard enthalpies of combustion ΔH_c° . [1]

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27 Calculating ΔH_f of Methane from Enthalpies of Combustion — 9701/22/O/N/21/Q5(a)–(b)

HARD · Syllabus 5.2

The standard enthalpy change of formation of methane, CH₄, cannot be determined directly.

Standard enthalpy changes of combustion:

- $\Delta H_c[\text{C}(\text{graphite})] = -393.5 \text{ kJ mol}^{-1}$
- $\Delta H_c[\text{H}_2(\text{g})] = -285.8 \text{ kJ mol}^{-1}$
- $\Delta H_c[\text{CH}_4(\text{g})] = -890.3 \text{ kJ mol}^{-1}$

- (a) Write the equation for the standard enthalpy change of formation of methane, CH₄(g). [1]

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- (b) Use the combustion data provided to calculate the standard enthalpy change of formation of methane, in kJ mol⁻¹. [3]

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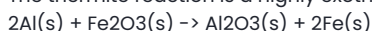
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28 Calculating ΔH_r from Enthalpies of Formation: Thermite Reaction — 9701/21/M/J/22/Q5(a)–(b)

EASY · Syllabus 5.2

The thermite reaction is a highly exothermic process used in railway track welding:



Standard enthalpies of formation:

- $\Delta H_f[\text{Fe}_2\text{O}_3(\text{s})] = -824.2 \text{ kJ mol}^{-1}$
- $\Delta H_f[\text{Al}_2\text{O}_3(\text{s})] = -1675.7 \text{ kJ mol}^{-1}$

- (a) State the value of ΔH_f^θ for Al(s) and Fe(s) . [1]
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- (b) Calculate the standard enthalpy change, ΔH_r^θ , for this reaction. [2]
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- (c) Explain why molten iron is produced by this reaction. [1]
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29 Enthalpy of Formation of Benzene from Combustion Data — 9701/22/M/J/22/Q4(a)–(b)

HARD · Syllabus 5.2

Standard enthalpy changes of combustion:

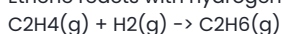
- $\text{C(s)} + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$ $\Delta H_c = -394 \text{ kJ mol}^{-1}$
- $\text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O(l)}$ $\Delta H_c = -286 \text{ kJ mol}^{-1}$
- $\text{C}_6\text{H}_6(\text{l}) + 7.5 \text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 3\text{H}_2\text{O(l)}$ $\Delta H_c = -3267 \text{ kJ mol}^{-1}$

- (a) Construct a Hess's Law cycle to calculate the standard enthalpy change of formation of liquid benzene, $\text{C}_6\text{H}_6(\text{l})$. [2]
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- (b) Calculate the standard enthalpy change of formation of liquid benzene in kJ mol^{-1} . [2]
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30 Bond Energy Calculations: Hydrogenation of Ethene — 9701/23/M/J/22/Q3(a)–(b)

EASY · Syllabus 5.2

Ethene reacts with hydrogen in the presence of a nickel catalyst to form ethane:



Relevant bond energies:

- $\text{C}=\text{C}$: 612 kJ mol^{-1}
- $\text{C}-\text{C}$: 348 kJ mol^{-1}
- $\text{C}-\text{H}$: 412 kJ mol^{-1}
- $\text{H}-\text{H}$: 436 kJ mol^{-1}

- (a) Calculate the total energy absorbed to break all necessary bonds in the reactants. [2]
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- (b) Calculate the total energy released when new bonds are formed in the product. [1]
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- (c) Calculate the enthalpy change of the reaction, ΔH , in kJ mol^{-1} . [1]
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31 Deducing an Unknown Bond Energy: C–H Bond in Methane — 9701/21/O/N/22/Q4(a)–(c)

HARD · Syllabus 5.2

The standard enthalpy changes of atomisation are:

- $\Delta H_{\text{at}}[\text{C(s)}] = +717 \text{ kJ mol}^{-1}$
- $\Delta H_{\text{at}}[\text{H}_2(\text{g})] = +218 \text{ kJ mol}^{-1}$ (per mole of H atoms)

The standard enthalpy change of formation of methane, $\Delta H_{\text{f}}[\text{CH}_4(\text{g})]$, is -75 kJ mol^{-1} .

- (a)** Write the equation for the complete atomisation of one mole of gaseous methane, $\text{CH}_4(\text{g})$. **[1]**

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- (b)** Construct a thermochemical cycle and calculate the total enthalpy of atomisation of $\text{CH}_4(\text{g})$. **[2]**

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- (c)** Calculate the mean C–H bond enthalpy in methane. **[1]**

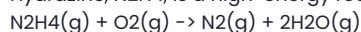
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32 Combustion of Hydrazine: Rocket Fuel Energetics — 9701/22/F/M/23/Q3(b)

HARD · Syllabus 5.2

Hydrazine, N_2H_4 , is a high-energy rocket fuel that reacts with oxygen:



Bond energies in kJ mol^{-1} :

- N–N: 158
- N=N: 418
- N#N (triple): 945
- N–H: 391
- O=O: 498
- O–H: 464

- (a)** Calculate the total energy absorbed to break all bonds in the reactants. **[2]**

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- (b)** Calculate the total energy released upon forming all bonds in the products. **[1]**

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- (c)** Calculate the enthalpy change of the reaction, ΔH , and explain why the reaction is extremely exothermic. **[2]**

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33 Why Bond Energy Calculations Differ from Standard Enthalpies — 9701/21/M/J/23/Q4(a)–(b)

HARD · Syllabus 5.2

The enthalpy change of combustion of gaseous ethanol calculated using mean bond energies is $-1270 \text{ kJ mol}^{-1}$, whereas the data book value obtained from standard enthalpies of formation is $-1367 \text{ kJ mol}^{-1}$.

- (a)** State two reasons why an enthalpy change calculated using bond energies differs from the value obtained from standard enthalpies of formation. **[2]**

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- (b) State whether bond energy calculations are applicable to reactions occurring in the liquid or solid states without modification, and explain why. [2]

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34 Hess's Law: Enthalpy of Formation of CS₂ — 9701/22/O/N/23/Q3(a)–(b)

HARD · Syllabus 5.2

Carbon disulfide, CS₂(l), burns completely according to the equation:



Standard enthalpy changes of combustion:

- $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) \quad \Delta H_{\text{c}} = -394 \text{ kJ mol}^{-1}$
- $\text{S}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{SO}_2(\text{g}) \quad \Delta H_{\text{c}} = -297 \text{ kJ mol}^{-1}$

- (a) Write the equation for the standard enthalpy change of formation of CS₂(l). [1]

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- (b) Calculate the standard enthalpy change of formation, $\Delta H_{\text{f}}^{\ominus}$, of CS₂(l) in kJ mol⁻¹. [3]

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35 Calculating Bond Energy of C=O in Carbon Dioxide — 9701/21/O/N/23/Q3(b)

HARD · Syllabus 5.2

Given the following thermochemical data:

- $\Delta H_{\text{at}}[\text{C}(\text{s})] = +717 \text{ kJ mol}^{-1}$
- Bond energy of O=O in O₂(g) = 498 kJ mol⁻¹
- $\Delta H_{\text{f}}^{\ominus}[\text{CO}_2(\text{g})] = -394 \text{ kJ mol}^{-1}$

- (a) Construct a thermochemical cycle connecting C(s) + O₂(g) to CO₂(g) and the separated gaseous atoms C(g) + 2O(g). [2]

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- (b) Calculate the bond energy of the C=O double bond in carbon dioxide. [2]

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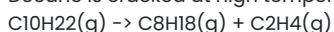
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36 Hess's Law: Enthalpy Change of Cracking Alkanes — 9701/22/M/J/24/Q4(a)–(b)

EASY · Syllabus 5.2

Decane is cracked at high temperature to produce octane and ethene:



Standard enthalpies of combustion:

- $\Delta H_{\text{c}}[\text{C}_{10}\text{H}_{22}(\text{g})] = -6778 \text{ kJ mol}^{-1}$
- $\Delta H_{\text{c}}[\text{C}_8\text{H}_{18}(\text{g})] = -5470 \text{ kJ mol}^{-1}$
- $\Delta H_{\text{c}}[\text{C}_2\text{H}_4(\text{g})] = -1411 \text{ kJ mol}^{-1}$

- (a) Calculate the enthalpy change, ΔH , for this cracking reaction. [2]

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- (b) State whether cracking is endothermic or exothermic, and explain why industrial cracking requires continuous high heat input. [2]

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37 Enthalpy of Formation of Gaseous Atoms: Nitrogen vs Oxygen — 9701/21/M/J/24/Q3(a)–(b)

HARD · Syllabus 5.2

The standard enthalpy changes of atomisation are:

- $\Delta H_{\text{at}}[1/2 \text{N}_2(\text{g}) \rightarrow \text{N}(\text{g})] = +473 \text{ kJ mol}^{-1}$
- $\Delta H_{\text{at}}[1/2 \text{O}_2(\text{g}) \rightarrow \text{O}(\text{g})] = +249 \text{ kJ mol}^{-1}$

- (a) Calculate the bond dissociation enthalpies of N_2 and O_2 . [2]

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- (b) Explain why the bond dissociation enthalpy of N_2 is nearly double that of O_2 . [2]

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38 Multi-Step Hess's Law Cycle: Phosphorus Chlorides — 9701/23/M/J/24/Q4(a)–(b)

HARD · Syllabus 5.2

Phosphorus reacts with chlorine in two stages to form PCl_3 and PCl_5 :

- $\text{P}_4(\text{s}) + 6\text{Cl}_2(\text{g}) \rightarrow 4\text{PCl}_3(\text{l})$ $\Delta H_1 = -1280 \text{ kJ mol}^{-1}$
- $\text{PCl}_3(\text{l}) + \text{Cl}_2(\text{g}) \rightarrow \text{PCl}_5(\text{s})$ $\Delta H_2 = -124 \text{ kJ mol}^{-1}$

- (a) Calculate the standard enthalpy change of formation of $\text{PCl}_3(\text{l})$. [1]

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- (b) Calculate the standard enthalpy change of formation of $\text{PCl}_5(\text{s})$. [2]

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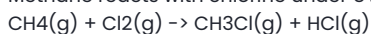
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39 Bond Enthalpy Calculation: Chlorination of Methane — 9701/22/O/N/24/Q4(a)–(c)

EASY · Syllabus 5.2

Methane reacts with chlorine under UV light to form chloromethane:



Bond energies in kJ mol^{-1} : C–H = 412, Cl–Cl = 242, C–Cl = 338, H–Cl = 431.

- (a) Identify which bonds are broken and which bonds are formed during this reaction. [2]

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- (b) Calculate the enthalpy change, ΔH , for this chlorination reaction. [2]

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40 Enthalpy of Combustion of Gaseous vs Liquid Alkanes — 9701/21/O/N/24/Q3(a)-(b)

HARD · Syllabus 5.2

Consider the combustion of hexane in gaseous and liquid states:

- $\text{C}_6\text{H}_{14}(\text{l}) + 9.5 \text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 7\text{H}_2\text{O}(\text{l})$ $\Delta H_{\text{c}} = -4163 \text{ kJ mol}^{-1}$
- The standard enthalpy change of vaporisation of hexane is $+31.5 \text{ kJ mol}^{-1}$.

- (a) State whether ΔH_{c} of gaseous hexane is more exothermic or less exothermic than ΔH_{c} of liquid hexane. [1]

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- (b) Calculate the standard enthalpy change of combustion of gaseous hexane, $\text{C}_6\text{H}_{14}(\text{g})$. [2]

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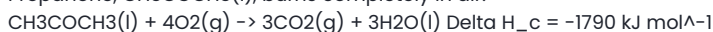
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41 Determining Enthalpy of Formation of Propanone — 9701/22/F/M/25/Q4(a)-(b)

HARD · Syllabus 5.2

Propanone, $\text{CH}_3\text{COCH}_3(\text{l})$, burns completely in air:



Standard enthalpies of formation:

- $\Delta H_{\text{f}}[\text{CO}_2(\text{g})] = -394 \text{ kJ mol}^{-1}$
- $\Delta H_{\text{f}}[\text{H}_2\text{O}(\text{l})] = -286 \text{ kJ mol}^{-1}$

- (a) Write the equation for the formation of propanone from its elements in their standard states. [1]

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- (b) Calculate the standard enthalpy change of formation, $\Delta H_{\text{f}}^{\theta}$, of propanone in kJ mol^{-1} . [3]

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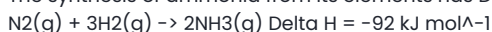
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42 Average Bond Energy of N-H in Ammonia — 9701/21/M/J/25/Q5(a)-(b)

HARD · Syllabus 5.2

The synthesis of ammonia from its elements has $\Delta H = -92 \text{ kJ mol}^{-1}$:



Bond energies: $E(\text{N}\equiv\text{N}) = 945 \text{ kJ mol}^{-1}$, $E(\text{H}-\text{H}) = 436 \text{ kJ mol}^{-1}$.

- (a) Calculate the total energy absorbed to break bonds in 1 mole of N_2 and 3 moles of H_2 . [1]

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- (b) Use the reaction enthalpy to calculate the mean N-H bond energy in ammonia. [3]

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43 Hess's Law: Enthalpy of Isomerisation of But-1-ene to But-2-ene — 9701/23/M/J/25/Q5(a)-(b)

EASY · Syllabus 5.2

But-1-ene isomerises to the more thermodynamically stable but-2-ene:

but-1-ene(g) → but-2-ene(g)

Standard enthalpies of combustion:

• $\Delta H_c[\text{but-1-ene(g)}] = -2719 \text{ kJ mol}^{-1}$

• $\Delta H_c[\text{but-2-ene(g)}] = -2710 \text{ kJ mol}^{-1}$

- (a) Calculate the enthalpy change of isomerisation, ΔH_{iso} . [2]

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- (b) Deduce which isomer is more stable and explain your reasoning in terms of enthalpy. [2]

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44 Deducing Bond Energy of Si-Si vs C-C — 9701/22/O/N/25/Q4(b)

EASY · Syllabus 5.2

Given the bond energies:

• C-C: 348 kJ mol^{-1}

• Si-Si: 226 kJ mol^{-1}

- (a) Explain why the silicon-silicon single bond is significantly weaker than the carbon-carbon single bond. [2]

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- (b) Relate this difference in bond energy to the relative thermal stability of alkanes compared to silanes (compounds of general formula $\text{Si}_n\text{H}_{2n+2}$). [2]

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45 Enthalpy of Formation of Gaseous Hydrogen Iodide: Bond Energy Anomaly — 9701/21/O/N/25/Q5(a)-(b)

HARD · Syllabus 5.2

For the formation of gaseous hydrogen iodide:

$\frac{1}{2} \text{H}_2(\text{g}) + \frac{1}{2} \text{I}_2(\text{s}) \rightarrow \text{HI}(\text{g})$ $\Delta H_{\text{f}} = +26.5 \text{ kJ mol}^{-1}$

Bond energies: $E(\text{H-H}) = 436 \text{ kJ mol}^{-1}$, $E(\text{H-I}) = 299 \text{ kJ mol}^{-1}$.

Standard enthalpy of sublimation of iodine: $\text{I}_2(\text{s}) \rightarrow \text{I}_2(\text{g})$ $\Delta H_{\text{sub}} = +62.4 \text{ kJ mol}^{-1}$.

- (a) Calculate the bond energy of the I-I bond in gaseous I_2 . [3]

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- (b) Explain why ΔH_{f} of $\text{HI}(\text{g})$ is endothermic even though bond formation occurs. [1]

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46 Enthalpy of Neutralisation: Dibasic vs Monobasic Acid — 9701/22/M/J/25/Q5(a)-(b)

HARD · Syllabus 5.2

Consider the two neutralisation reactions:

Reaction 1: $\text{HCl(aq)} + \text{NaOH(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$ $\Delta H_1 = -57.1 \text{ kJ mol}^{-1}$

Reaction 2: $(\text{COOH})_2(\text{aq}) + 2\text{NaOH(aq)} \rightarrow (\text{COONa})_2(\text{aq}) + 2\text{H}_2\text{O(l)}$ $\Delta H_2 = -106.4 \text{ kJ mol}^{-1}$

- (a)** Calculate the enthalpy change of neutralisation per mole of water formed in Reaction 2. **[1]**

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- (b)** Explain why the enthalpy change per mole of water in Reaction 2 is less exothermic than $-57.1 \text{ kJ mol}^{-1}$. **[2]**

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47 Bond Energy of Carbon-Carbon Triple Bond in Ethyne — 9701/23/O/N/25/Q5(a)-(b)

HARD · Syllabus 5.2

The complete combustion of ethyne gas, $\text{C}_2\text{H}_2(\text{g})$, has $\Delta H_c = -1300 \text{ kJ mol}^{-1}$:

$\text{C}_2\text{H}_2(\text{g}) + 2.5 \text{ O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$

Bond energies in kJ mol^{-1} :

- C-H: 412
- O=O: 498
- C=O (in CO_2): 805
- O-H (in H_2O): 464

- (a)** Calculate the total energy released upon forming all bonds in 2 moles of $\text{CO}_2(\text{g})$ and 1 mole of $\text{H}_2\text{O}(\text{g})$. **[2]**

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- (b)** Calculate the bond energy of the carbon-carbon triple bond ($\text{C}\equiv\text{C}$) in ethyne. **[3]**

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48 Lattice Energy and Enthalpy of Solution of Potassium Fluoride — 9701/21/F/M/24/Q5(a)-(b)

HARD · Syllabus 5.2

The enthalpy of solution of potassium fluoride, KF , is $-17.7 \text{ kJ mol}^{-1}$.

- Enthalpy of hydration of $\text{K}^+(\text{g}) = -322 \text{ kJ mol}^{-1}$
- Enthalpy of hydration of $\text{F}^-(\text{g}) = -506 \text{ kJ mol}^{-1}$

- (a)** State the relationship connecting enthalpy of solution, lattice energy (endothermic dissociation), and hydration enthalpies. **[1]**

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- (b)** Calculate the lattice energy of potassium fluoride (defined as $\text{KF(s)} \rightarrow \text{K}^+(\text{g}) + \text{F}^-(\text{g})$). **[2]**

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49 Hess's Law: Enthalpy of Formation of Nitric Acid — 9701/22/M/J/24/Q5(a)–(b)

HARD · Syllabus 5.2

In the Ostwald process, nitrogen dioxide reacts with water to produce nitric acid:



Standard enthalpies of formation:

- $\Delta H_{\text{f}}^{\ominus}[\text{NO}_2(\text{g})] = +33.2 \text{ kJ mol}^{-1}$
- $\Delta H_{\text{f}}^{\ominus}[\text{H}_2\text{O}(\text{l})] = -285.8 \text{ kJ mol}^{-1}$
- $\Delta H_{\text{f}}^{\ominus}[\text{NO}(\text{g})] = +90.2 \text{ kJ mol}^{-1}$

- (a) Construct an expression for ΔH_{r} in terms of standard enthalpies of formation. [1]

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- (b) Calculate the standard enthalpy change of formation of aqueous nitric acid, $\text{HNO}_3(\text{aq})$. [3]

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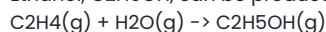
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50 Comprehensive Synoptic: Hess's Law, Calorimetry & Bond Energies of Ethanol — 9701/22/O/N/25/Q5(a)–(d) HARD · Syllabus 5.2

Ethanol, $\text{C}_2\text{H}_5\text{OH}$, can be produced industrially by the direct catalytic hydration of ethene:



Standard enthalpies of combustion:

- $\Delta H_{\text{c}}[\text{C}_2\text{H}_4(\text{g})] = -1411 \text{ kJ mol}^{-1}$
- $\Delta H_{\text{c}}[\text{C}_2\text{H}_5\text{OH}(\text{l})] = -1367 \text{ kJ mol}^{-1}$
- Enthalpy of vaporisation of ethanol = $+42 \text{ kJ mol}^{-1}$

Bond energies in kJ mol^{-1} : $\text{C}=\text{C} = 612$, $\text{C}-\text{C} = 348$, $\text{C}-\text{H} = 412$, $\text{C}-\text{O} = 360$, $\text{O}-\text{H} = 464$.

- (a) Use the bond energies provided to calculate the enthalpy change for the hydration of ethene, $\Delta H_{\text{hydration}}$. [3]

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- (b) Use the combustion data and enthalpy of vaporisation to calculate an alternative value for $\Delta H_{\text{hydration}}$. [3]

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- (c) Explain why the value obtained from bond energies in (a) differs from the value obtained from combustion data in (b). [2]

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- (d) State whether high temperature or low temperature favours a high equilibrium yield of ethanol in this hydration reaction, justifying your answer using Le Chatelier's principle. [2]

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Mark Scheme — Topic 5 — Chemical Energetics

CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	The enthalpy change when one mole of a compound (1); is formed from its constituent elements in their standard states (1); under standard conditions (298 K and 100 kPa) (1)	3
1(b)	Temperature: 298 K (or 25 °C) (1); Pressure: 100 kPa (or 1.00 x 10 ⁵ Pa / 1 bar) (1)	2
1(c)	Bromine is an element in its standard state at 298 K and 100 kPa (1)	1
2(a)	The minimum energy required by colliding particles to initiate a chemical reaction (1)	1
2(b)	A catalyst provides an alternative reaction pathway with a lower activation energy, E _a ' (1); The energy levels of initial reactants and final products remain unchanged, so Delta H is unchanged (1)	2
2(c)	Delta H = E _a (forward) - E _a (reverse) (or E _a (forward) = E _a (reverse) + Delta H) (1)	1
3(a)	The enthalpy change when one mole of a substance (1); is completely burned in excess oxygen (1); under standard conditions (298 K and 100 kPa) with all reactants and products in standard states (1)	3
3(b)	C ₂ H ₅ OH(l) + 3O ₂ (g) → 2CO ₂ (g) + 3H ₂ O(l) (1); Correct state symbols throughout (l, g, g, l) (1)	2
4(a)	The enthalpy change when one mole of water is formed (1); from the reaction between an acid and an alkali / base under standard conditions (1)	2
4(b)	H ⁺ (aq) + OH ⁻ (aq) → H ₂ O(l) (1)	1
4(c)	Ethanoic acid is a weak acid that is only partially dissociated in aqueous solution (1); Some energy is absorbed (endothermic step) to ionise unreacted CH ₃ COOH molecules fully during neutralisation, making the overall process less exothermic (1)	2
5(a)	Delta T = 44.0 - 19.5 = 24.5 °C (or K); q = mc Delta T = 150.0 x 4.18 x 24.5 = 15361 J (1); q = 15.4 kJ (1)	2
5(b)	Mass burned = 78.45 - 77.85 = 0.60 g; n = 0.60 / 60.0 = 0.0100 mol (1); Delta H _c = -q / n = -15.361 / 0.0100 = -1536 kJ mol ⁻¹ (allow -1530 to -1540 kJ mol ⁻¹) (1)	2
5(c)	Significant heat loss to the surrounding air / draughts / copper can (1); Incomplete combustion of propan-1-ol (forming soot/CO instead of CO ₂) / evaporation of alcohol from wick (1)	2
6(a)	Extrapolated maximum temperature at t = 3.0 min = 34.7 °C (allow 34.5 - 34.9 °C) (1); Delta T _{corrected} = 34.7 - 20.0 = 14.7 °C (allow 14.5 - 14.9 °C) (1)	2
6(b)	m = 50.0 g; q = mc Delta T = 50.0 x 4.18 x 14.7 = 3072 J = 3.072 kJ (1); n(CuSO ₄) = c x V = 1.00 x 0.0500 = 0.0500 mol (1); Delta H = -q / n = -3.072 / 0.0500 = -61.4 kJ mol ⁻¹ (allow -60.6 to -62.3 kJ mol ⁻¹) (1)	3
6(c)	Polystyrene is a good thermal insulator / has a very low heat capacity, minimising heat loss to surroundings (1)	1
7(a)	Total mass m = 25.0 + 50.0 = 75.0 g; Delta T = 29.8 - 21.0 = 8.8 °C (1); q = 75.0 x 4.18 x 8.8 = 2758.8 J = 2.76 kJ (1)	2
7(b)	Moles of H ₂ O formed: n(H ₂ SO ₄) = 0.0250 mol, n(KOH) = 0.0500 mol ⇒ n(H ₂ O) = 0.0500 mol (1); Delta H _{neut} = -2.759 / 0.0500 = -55.2 kJ mol ⁻¹ (allow -55.0 to -55.5 kJ mol ⁻¹) (1)	2
8(a)	Endothermic (temperature decreases) (1)	1
8(b)	Delta T = 16.2 - 20.0 = -3.8 °C; q = 100.0 x 4.18 x 3.8 = 1588.4 J = 1.588 kJ (1); n(NH ₄ Cl) = 5.35 / 53.5 = 0.100 mol (1); Delta H _{sol} = +1.588 / 0.100 = +15.9 kJ mol ⁻¹ (1)	3
8(c)	Horizontal line for reactants NH ₄ Cl(s) + aq lower than products NH ₄ ⁺ (aq) + Cl ⁻ (aq) (1); Upward arrow labelled Delta H = +15.9 kJ mol ⁻¹ (1)	2
9(a)	The enthalpy change when one mole of gaseous atoms (1); is formed from the element in its standard state (1); under standard conditions (298 K and 100 kPa) (1)	3
9(b)	1/2 Br ₂ (l) → Br(g) (1)	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
9(c)	Equation for atomisation: $\frac{1}{2} \text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$; Equation for bond dissociation: $\text{Cl}_2(\text{g}) \rightarrow 2\text{Cl}(\text{g})$ (1); Therefore $\Delta H_{\text{at}}(\text{Cl}_2) = \frac{1}{2} E(\text{Cl}-\text{Cl})$ (1)	2
10(a)	The enthalpy change when one mole of specified gaseous ions (1); dissolves in sufficient water to form an infinitely dilute solution under standard conditions (1)	2
10(b)	Mg^{2+} carries a 2+ charge compared to 1+ for Na^+ (1); Mg^{2+} has a significantly smaller ionic radius than Na^+ (1); Mg^{2+} has a much higher charge density, forming substantially stronger ion-dipole electrostatic attractions with water molecules (1)	3
11(a)	$q_{\text{water}} = 200.0 \times 4.18 \times 12.4 = 10366.4 \text{ J} = 10.37 \text{ kJ}$ (1); $q_{\text{cal}} = 85.0 \times 12.4 = 1054.0 \text{ J} = 1.05 \text{ kJ}$ (1); $q_{\text{total}} = 10366.4 + 1054.0 = 11420.4 \text{ J} = 11.42 \text{ kJ}$ (1)	3
11(b)	Difference = 1054 J; Percentage error = $(1054 / 11420) \times 100\% = 9.23\%$ (allow 9.2% – 9.3%) (1); The calculated enthalpy would be 9.2% too low / less exothermic (1)	2
12(a)	Enthalpies of combustion become increasingly negative / more exothermic (1); Each successive alkane has an additional $-\text{CH}_2-$ group, requiring more O_2 and forming more $\text{C}=\text{O}$ and $\text{O}-\text{H}$ bonds which release more energy upon formation (1)	2
12(b)	Average increment per $\text{CH}_2 = (2877 - 890) / 3 = 662 \text{ kJ mol}^{-1}$ (1); Estimated $\Delta H_{\text{c}}(\text{pentane}) = -2877 - 660 = -3537 \text{ kJ mol}^{-1}$ (allow -3530 to $-3550 \text{ kJ mol}^{-1}$) (1)	2
13(a)	Endothermic (1); Energy must be absorbed / supplied from surroundings to overcome the electrostatic attraction holding the bonded atoms together (1)	2
13(b)	More energy is released when new bonds are formed in the products (1); than the energy absorbed to break the original bonds in the reactants (1)	2
14(a)	$n(\text{Cu}) = 0.635 / 63.5 = 0.0100 \text{ mol}$; $n(\text{AgNO}_3) = 0.200 \times 0.100 = 0.0200 \text{ mol}$ (1); Stoichiometric ratio is 1 Cu : 2 AgNO ₃ ; exactly equal stoichiometric amounts, both react completely (or Cu is limiting) (1)	2
14(b)	$q = mc \Delta T = 100.0 \times 4.18 \times 3.5 = 1463 \text{ J} = 1.463 \text{ kJ}$ (1); $n(\text{Cu}) = 0.0100 \text{ mol}$ (1); $\Delta H = -q / n = -1.463 / 0.0100 = -146.3 \text{ kJ mol}^{-1}$ (allow -146 kJ mol^{-1}) (1)	3
15(a)	$m = 100.0 \text{ g}$; $q = 100.0 \times 4.18 \times 6.1 = 2549.8 \text{ J} = 2.55 \text{ kJ}$ (1); Correct sign and units (1)	2
15(b)	$n = 0.0500 \text{ mol}$; $\Delta H_{\text{neut}} = -2.550 / 0.0500 = -51.0 \text{ kJ mol}^{-1}$ (allow -50.8 to $-51.2 \text{ kJ mol}^{-1}$) (1); Correct negative sign (1)	2
15(c)	Aqueous ammonia is a weak base and only partially ionised (1); Some heat is absorbed to ionise $\text{NH}_3(\text{aq})$ molecules into $\text{NH}_4^+(\text{aq})$ and $\text{OH}^-(\text{aq})$ during the reaction (1)	2
16(a)	Under standard conditions (298 K and 100 kPa), the standard state of water is liquid, $\text{H}_2\text{O}(\text{l})$, not gas, $\text{H}_2\text{O}(\text{g})$ (1); Standard enthalpy of formation requires products to be in their standard states (1)	2
16(b)	$\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{g})$ $\Delta H_{\text{vap}} = +44.0 \text{ kJ mol}^{-1}$ (1); $\Delta H_{\text{f}}(\text{H}_2\text{O}(\text{l})) = -241.8 - (+44.0) = -285.8 \text{ kJ mol}^{-1}$ (1)	2
17(a)	Heat is absorbed from the surroundings (1)	1
17(b)	Mass of $\text{CaCO}_3 = 2.50 \times 10^6 \text{ g}$; $n(\text{CaCO}_3) = (2.50 \times 10^6) / 100.1 = 24975 \text{ mol}$ (1); Energy required = $24975 \times 178 \text{ kJ} = 4.446 \times 10^6 \text{ kJ}$ (1); Energy in MJ = 4446 MJ (or $4.45 \times 10^3 \text{ MJ}$) (1)	3
18(a)	$n(\text{H}_2\text{SO}_4) = 0.0500 \text{ mol}$, $n(\text{NaOH}) = 0.100 \text{ mol} \Rightarrow n(\text{H}_2\text{O}) = 0.100 \text{ mol}$ (1)	1
18(b)	$m = 150.0 \text{ g}$; $q = 150.0 \times 4.18 \times 9.1 = 5705.7 \text{ J} = 5.71 \text{ kJ}$ (1); Correct units (1)	2
18(c)	$\Delta H_{\text{neut}} = -5.706 / 0.100 = -57.1 \text{ kJ mol}^{-1}$ (1)	1
19(a)	The average energy required to break one mole of a specific covalent bond (1); in gaseous molecules (1); averaged over a wide range of different chemical compounds (1)	3
19(b)	Bond breaking is always an endothermic process / requires energy input to overcome attractions (1)	1
20(a)	Energy per gram = $1367 / 46.0 = 29.7 \text{ kJ g}^{-1}$ (1); Correct calculation to 3 s.f. (1)	2
20(b)	Energy per gram = $5480 / 158.0 = 34.7 \text{ kJ g}^{-1}$ (1)	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
20(c)	Biofuels are renewable / carbon-neutral as crops absorb CO ₂ during photosynthesis equal to CO ₂ released upon combustion (1)	1
21(a)	$q = 250 \times 4.18 \times 35.0 = 36575 \text{ J} = 36.575 \text{ kJ}$ (1); $n(\text{CaCl}_2) \text{ required} = 36.575 / 82.8 = 0.4417 \text{ mol}$ (1); Mass = $0.4417 \times 111.1 = 49.1 \text{ g}$ (allow 48.9 – 49.3 g) (1)	3
21(b)	$\Delta H_{\text{sol}} = \text{sum}(\Delta H_{\text{hyd}}) - (\text{Lattice Energy})$ (1); For CaCl ₂ , the exothermic hydration enthalpy of Ca ²⁺ and Cl ⁻ exceeds the endothermic lattice energy (sum ΔH_{hyd} is more exothermic than lattice energy is endothermic) (1)	2
22(a)	Use a draught shield / windshield around burner and calorimeter (1); Place an insulating lid on the copper calorimeter / use a bomb calorimeter with pure oxygen supply (1)	2
22(b)	Incomplete combustion produces carbon monoxide (CO) or soot (C) rather than CO ₂ (1); The combustion of carbon to CO or C releases significantly less energy than complete combustion to CO ₂ (1)	2
23(a)	When carbon burns in a limited supply of oxygen, a mixture of both CO and CO ₂ is inevitably formed (1); Unreacted carbon may also remain, so the reaction cannot be stopped purely at CO (1)	2
23(b)	Construct a thermochemical cycle using the readily measurable enthalpies of combustion of carbon (to CO ₂) and carbon monoxide (to CO ₂) (1); Apply Hess's Law: $\Delta H_{\text{f}}(\text{CO}) = \Delta H_{\text{c}}(\text{C}) - \Delta H_{\text{c}}(\text{CO})$ (1)	2
24(a)	Endothermic, $\Delta H = +196 \text{ kJ mol}^{-1}$ (1)	1
24(b)	$E_{\text{a}}(\text{reverse}) = E_{\text{a}}(\text{forward}) - \Delta H$ (1); $E_{\text{a}}(\text{reverse}) = +65 - (-196) = +261 \text{ kJ mol}^{-1}$ (1)	2
25(a)	$\text{CuSO}_4(\text{s}) + 5\text{H}_2\text{O}(\text{l}) \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$ with $\Delta H_{\text{hydration}}$ (1); Both react with excess water to form $\text{CuSO}_4(\text{aq})$ via ΔH_1 and ΔH_2 (1)	2
25(b)	$\Delta H_{\text{hydration}} + \Delta H_2 = \Delta H_1 \Rightarrow \Delta H_{\text{hydration}} = \Delta H_1 - \Delta H_2$ (1); $\Delta H_{\text{hydration}} = -66.5 - (+11.7) = -78.2 \text{ kJ mol}^{-1}$ (1)	2
26(a)	The total enthalpy change for a chemical reaction is independent of the route taken (1); provided the initial and final states are the same (1)	2
26(b)	$\Delta H_{\text{r}} = \text{sum } \Delta H_{\text{f}}(\text{products}) - \text{sum } \Delta H_{\text{f}}(\text{reactants})$ (1)	1
26(c)	$\Delta H_{\text{r}} = \text{sum } \Delta H_{\text{c}}(\text{reactants}) - \text{sum } \Delta H_{\text{c}}(\text{products})$ (1)	1
27(a)	$\text{C}(\text{graphite}) + 2\text{H}_2(\text{g}) \rightarrow \text{CH}_4(\text{g})$ (1)	1
27(b)	$\Delta H_{\text{f}} = \text{sum } \Delta H_{\text{c}}(\text{reactants}) - \text{sum } \Delta H_{\text{c}}(\text{products})$ (1); $\Delta H_{\text{f}} = [\Delta H_{\text{c}}(\text{C}) + 2 \times \Delta H_{\text{c}}(\text{H}_2)] - [\Delta H_{\text{c}}(\text{CH}_4)] = [-393.5 + 2(-285.8)] - [-890.3]$ (1); $\Delta H_{\text{f}} = -965.1 - (-890.3) = -74.8 \text{ kJ mol}^{-1}$ (1)	3
28(a)	0.0 kJ mol^{-1} (for both elements in their standard states) (1)	1
28(b)	$\Delta H_{\text{r}} = \Delta H_{\text{f}}(\text{Al}_2\text{O}_3) - \Delta H_{\text{f}}(\text{Fe}_2\text{O}_3) = -1675.7 - (-824.2)$ (1); $\Delta H_{\text{r}} = -851.5 \text{ kJ mol}^{-1}$ (1)	2
28(c)	The enormous heat energy released raises the temperature of the reaction mixture above the melting point of iron (1538 °C) (1)	1
29(a)	Cycle showing $6\text{C}(\text{s}) + 3\text{H}_2(\text{g}) \rightarrow \text{C}_6\text{H}_6(\text{l})$ at top (1); Downward arrows from reactants and product to combustion products $6\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l})$ (1)	2
29(b)	$\Delta H_{\text{f}} = [6(-394) + 3(-286)] - [-3267] = [-2364 - 858] - [-3267]$ (1); $\Delta H_{\text{f}} = -3222 + 3267 = +45 \text{ kJ mol}^{-1}$ (allow +45 to +49 kJ mol ⁻¹) (1)	2
30(a)	Bonds broken: $1 \times (\text{C}=\text{C}) + 1 \times (\text{H}-\text{H}) = 612 + 436 = 1048 \text{ kJ}$ (or including 4 C-H: $612 + 436 + 4(412) = 2696 \text{ kJ}$) (1); Correct summation (1)	2
30(b)	Bonds formed: $1 \times (\text{C}-\text{C}) + 2 \times (\text{C}-\text{H}) = 348 + 2(412) = 1172 \text{ kJ}$ (or including all 6 C-H: $348 + 6(412) = 2820 \text{ kJ}$) (1)	1
30(c)	$\Delta H = \text{Bonds Broken} - \text{Bonds Formed} = 1048 - 1172 = -124 \text{ kJ mol}^{-1}$ (1)	1
31(a)	$\text{CH}_4(\text{g}) \rightarrow \text{C}(\text{g}) + 4\text{H}(\text{g})$ (1)	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
31(b)	$\Delta H_{\text{atomisation}}(\text{CH}_4) = -\Delta H_{\text{f}}(\text{CH}_4) + \Delta H_{\text{at}}(\text{C}) + 4 \times \Delta H_{\text{at}}(\text{H})$ (1); $= -(-75) + 717 + 4(218) = 75 + 717 + 872 = +1664 \text{ kJ mol}^{-1}$ (1)	2
31(c)	Mean $E(\text{C-H}) = 1664 / 4 = 416 \text{ kJ mol}^{-1}$ (1)	1
32(a)	Bonds broken = $1(\text{N-N}) + 4(\text{N-H}) + 1(\text{O=O}) = 158 + 4(391) + 498 = 158 + 1564 + 498 = 2220 \text{ kJ mol}^{-1}$ (1); Correct working (1)	2
32(b)	Bonds formed = $1(\text{N}\equiv\text{N}) + 4(\text{O-H}) = 945 + 4(464) = 945 + 1856 = 2801 \text{ kJ mol}^{-1}$ (1)	1
32(c)	$\Delta H = 2220 - 2801 = -581 \text{ kJ mol}^{-1}$ (1); The reaction is extremely exothermic because of the formation of the exceptionally strong nitrogen-nitrogen triple bond ($\text{N}\equiv\text{N}$, 945 kJ mol^{-1}) and strong O-H bonds (1)	2
33(a)	Bond energies are average values taken across a wide variety of different molecules with different chemical environments (1); Standard enthalpies of formation use specific values determined under standard conditions (298 K, 100 kPa) where ethanol and water are liquids, whereas bond energies apply strictly to gaseous species (1)	2
33(b)	No (1); Bond energies assume all reactants and products are in the gaseous state with zero intermolecular forces; for liquids and solids, enthalpy changes of vaporisation/fusion must also be accounted for (1)	2
34(a)	$\text{C(s)} + 2\text{S(s)} \rightarrow \text{CS}_2(\text{l})$ (1)	1
34(b)	$\Delta H_{\text{f}} = [\Delta H_{\text{c}}(\text{C}) + 2 \times \Delta H_{\text{c}}(\text{S})] - [\Delta H_{\text{c}}(\text{CS}_2)]$ (1); $= [-394 + 2(-297)] - [-1075] = [-394 - 594] - [-1075] = -988 + 1075$ (1); $\Delta H_{\text{f}} = +87 \text{ kJ mol}^{-1}$ (1)	3
35(a)	Cycle showing formation route from elements to CO_2 ($\Delta H_{\text{f}} = -394$) and atomisation route to $\text{C(g)} + 2\text{O(g)}$ (1); Atomisation of reactants = $\Delta H_{\text{at}}(\text{C}) + E(\text{O=O}) = 717 + 498 = +1215 \text{ kJ mol}^{-1}$ (1)	2
35(b)	Atomisation of CO_2 : $2 \times E(\text{C=O}) = 1215 - (-394) = 1609 \text{ kJ mol}^{-1}$ (1); $E(\text{C=O}) = 1609 / 2 = 805 \text{ kJ mol}^{-1}$ (allow 804 - 806 kJ mol^{-1}) (1)	2
36(a)	$\Delta H = \Delta H_{\text{c}}(\text{C}_{10}\text{H}_{22}) - [\Delta H_{\text{c}}(\text{C}_8\text{H}_{18}) + \Delta H_{\text{c}}(\text{C}_2\text{H}_4)] = -6778 - [-5470 + (-1411)]$ (1); $\Delta H = -6778 - (-6881) = +103 \text{ kJ mol}^{-1}$ (1)	2
36(b)	Endothermic (ΔH is positive) (1); Thermal energy is continuously required to break strong C-C covalent bonds in long hydrocarbon chains (1)	2
37(a)	$E(\text{N}\equiv\text{N}) = 2 \times 473 = 946 \text{ kJ mol}^{-1}$ (1); $E(\text{O=O}) = 2 \times 249 = 498 \text{ kJ mol}^{-1}$ (1)	2
37(b)	N_2 possesses a triple covalent bond (one sigma and two pi bonds) shared between two nitrogen atoms (1); O_2 possesses only a double covalent bond (one sigma and one pi bond), which requires significantly less energy to break (1)	2
38(a)	$\Delta H_{\text{f}}(\text{PCl}_3) = -1280 / 4 = -320 \text{ kJ mol}^{-1}$ (1)	1
38(b)	Formation of PCl_5 : $1/4 \text{ P}_4(\text{s}) + 5/2 \text{ Cl}_2(\text{g}) \rightarrow \text{PCl}_5(\text{s})$ (1); $\Delta H_{\text{f}}(\text{PCl}_5) = \Delta H_{\text{f}}(\text{PCl}_3) + \Delta H_{\text{2}} = -320 + (-124) = -444 \text{ kJ mol}^{-1}$ (1)	2
39(a)	Bonds broken: one C-H bond and one Cl-Cl bond (1); Bonds formed: one C-Cl bond and one H-Cl bond (1)	2
39(b)	Bonds broken = $412 + 242 = 654 \text{ kJ mol}^{-1}$; Bonds formed = $338 + 431 = 769 \text{ kJ mol}^{-1}$ (1); $\Delta H = 654 - 769 = -115 \text{ kJ mol}^{-1}$ (1)	2
40(a)	More exothermic (ΔH is more negative) (1)	1
40(b)	$\text{C}_6\text{H}_{14}(\text{l}) \rightarrow \text{C}_6\text{H}_{14}(\text{g})$ $\Delta H_{\text{vap}} = +31.5 \text{ kJ mol}^{-1}$; $\Delta H_{\text{c}}(\text{liquid}) = \Delta H_{\text{vap}} + \Delta H_{\text{c}}(\text{gas})$ (1); $\Delta H_{\text{c}}(\text{gas}) = -4163 - 31.5 = -4194.5 \text{ kJ mol}^{-1}$ (allow $-4195 \text{ kJ mol}^{-1}$) (1)	2
41(a)	$3\text{C(s)} + 3\text{H}_2(\text{g}) + 1/2 \text{ O}_2(\text{g}) \rightarrow \text{CH}_3\text{COCH}_3(\text{l})$ (1)	1
41(b)	$\Delta H_{\text{c}} = [3 \times \Delta H_{\text{f}}(\text{CO}_2) + 3 \times \Delta H_{\text{f}}(\text{H}_2\text{O})] - \Delta H_{\text{f}}(\text{propanone})$ (1); $-1790 = [3(-394) + 3(-286)] - \Delta H_{\text{f}}(\text{propanone})$ (1); $\Delta H_{\text{f}}(\text{propanone}) = [-1182 - 858] - \Delta H_{\text{f}} = -2040 - \Delta H_{\text{f}}$ (1); $\Delta H_{\text{f}} = -2040 - (-1790) = -250 \text{ kJ mol}^{-1}$ (1)	3
42(a)	Bonds broken = $945 + 3(436) = 945 + 1308 = 2253 \text{ kJ}$ (1)	1
42(b)	$\Delta H = \text{Bonds broken} - \text{Bonds formed} = -92 = 2253 - 6 \times E(\text{N-H})$ (1); $6 \times E(\text{N-H}) = 2253 - (-92) = 2345 \text{ kJ}$ (1); $E(\text{N-H}) = 2345 / 6 = 390.8 \text{ kJ mol}^{-1}$ (allow 391 kJ mol^{-1}) (1)	3

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
43(a)	$\Delta H_{\text{iso}} = \Delta H_{\text{c}}(\text{but-1-ene}) - \Delta H_{\text{c}}(\text{but-2-ene}) = -2719 - (-2710) \text{ (1)}; \Delta H_{\text{iso}} = -9 \text{ kJ mol}^{-1} \text{ (1)}$	2
43(b)	But-2-ene is more stable (1); It is at a lower enthalpy level / releases less energy upon combustion (1)	2
44(a)	Silicon atoms have a larger atomic radius than carbon atoms (3 shells vs 2 shells) (1); The shared electron pair is further from the silicon nuclei and shielded by inner electron shells, resulting in poorer orbital overlap and weaker electrostatic attraction (1)	2
44(b)	Alkanes are thermally stable due to strong C-C and C-H bonds (1); Silanes decompose and ignite spontaneously in air because Si-Si and Si-H bonds have significantly lower bond enthalpies (1)	2
45(a)	Hess's route: $\Delta H_{\text{f}} = 1/2 E(\text{H-H}) + 1/2 \Delta H_{\text{sub}}(\text{I}_2) + 1/2 E(\text{I-I}) - E(\text{H-I}) \text{ (1)}; +26.5 = 1/2(436) + 1/2(62.4) + 1/2 E(\text{I-I}) - 299 = 218 + 31.2 + 1/2 E(\text{I-I}) - 299 = -49.8 + 1/2 E(\text{I-I}) \text{ (1)}; 1/2 E(\text{I-I}) = 26.5 + 49.8 = 76.3 \Rightarrow E(\text{I-I}) = 152.6 \text{ kJ mol}^{-1} \text{ (allow 151 - 154 kJ mol}^{-1}) \text{ (1)}$	3
45(b)	The energy released by forming the relatively weak H-I bond is insufficient to compensate for the energy required to atomise H_2 and sublime/dissociate I_2 (1)	1
46(a)	ΔH per mole of water = $-106.4 / 2 = -53.2 \text{ kJ mol}^{-1} \text{ (1)}$	1
46(b)	Ethanedioic acid (oxalic acid) is a weak acid that is only partially dissociated into ions in solution (1); Energy is taken in / absorbed (endothermic) to break O-H bonds and fully dissociate the acid molecules during reaction (1)	2
47(a)	Bonds formed = $4 \times E(\text{C=O}) + 2 \times E(\text{O-H}) = 4(805) + 2(464) \text{ (1)}; = 3220 + 928 = 4148 \text{ kJ mol}^{-1} \text{ (1)}$	2
47(b)	$\Delta H = \text{Bonds broken} - \text{Bonds formed} \Rightarrow -1300 = [E(\text{C}\equiv\text{C}) + 2(412) + 2.5(498)] - 4148 \text{ (1)}; -1300 = E(\text{C}\equiv\text{C}) + 824 + 1245 - 4148 = E(\text{C}\equiv\text{C}) - 2079 \text{ (1)}; E(\text{C}\equiv\text{C}) = 2079 - 1300 = 779 \text{ kJ mol}^{-1} \text{ (allow 775 - 840 kJ mol}^{-1} \text{ depending on standard C=O value) (1)}$	3
48(a)	$\Delta H_{\text{sol}} = \text{Lattice dissociation energy} + \text{sum } \Delta H_{\text{hyd}}$ (or $\Delta H_{\text{sol}} = \text{sum } \Delta H_{\text{hyd}} - \text{Lattice formation energy}$) (1)	1
48(b)	Total $\Delta H_{\text{hyd}} = -322 + (-506) = -828 \text{ kJ mol}^{-1} \text{ (1)}; -17.7 = \text{Lattice dissociation energy} + (-828) \Rightarrow \text{Lattice dissociation energy} = 828 - 17.7 = +810.3 \text{ kJ mol}^{-1} \text{ (1)}$	2
49(a)	$\Delta H_{\text{r}} = [2 \times \Delta H_{\text{f}}(\text{HNO}_3) + \Delta H_{\text{f}}(\text{NO})] - [3 \times \Delta H_{\text{f}}(\text{NO}_2) + \Delta H_{\text{f}}(\text{H}_2\text{O})] \text{ (1)}$	1
49(b)	$-138 = [2 \times \Delta H_{\text{f}}(\text{HNO}_3) + 90.2] - [3(33.2) + (-285.8)] = [2 \times \Delta H_{\text{f}}(\text{HNO}_3) + 90.2] - [99.6 - 285.8] \text{ (1)}; -138 = 2 \times \Delta H_{\text{f}}(\text{HNO}_3) + 90.2 - (-186.2) = 2 \times \Delta H_{\text{f}}(\text{HNO}_3) + 276.4 \text{ (1)}; 2 \times \Delta H_{\text{f}}(\text{HNO}_3) = -138 - 276.4 = -414.4 \Rightarrow \Delta H_{\text{f}}(\text{HNO}_3) = -207.2 \text{ kJ mol}^{-1} \text{ (1)}$	3
50(a)	Bonds broken = $1(\text{C}\equiv\text{C}) + 1(\text{O-H}) = 612 + 464 = 1076 \text{ kJ}$; Bonds formed = $1(\text{C-C}) + 1(\text{C-H}) + 1(\text{C-O}) = 348 + 412 + 360 = 1120 \text{ kJ}$ (1); $\Delta H = 1076 - 1120 = -44 \text{ kJ mol}^{-1} \text{ (allow -42 to -46 kJ mol}^{-1}) \text{ (2)}$	3
50(b)	$\Delta H_{\text{c}}(\text{ethanol gas}) = -1367 - (+42) = -1409 \text{ kJ mol}^{-1} \text{ (1)}; \Delta H_{\text{hydration}} = \Delta H_{\text{c}}(\text{C}_2\text{H}_4) - \Delta H_{\text{c}}(\text{ethanol gas}) = -1411 - (-1409) = -2 \text{ kJ mol}^{-1} \text{ (or using liquid: -1411 - (-1367) = -44 kJ mol}^{-1}) \text{ (2)}$	3
50(c)	Mean bond energies are average values across various molecular environments rather than specific to ethanol (1); Real reaction involves specific molecular geometries, bond polarities, and slight non-ideality (1)	2
50(d)	Low temperature favours a higher yield (1); The forward reaction is exothermic ($\Delta H < 0$), so lowering temperature shifts equilibrium in the exothermic forward direction to release heat (Le Chatelier's principle) (1)	2